UTILIZATION OF THE SAME DATA FOR ADVANCED MODELING AND SIMULATION OF SMOKE GENERATION IN MICROGRAVITY

Sean C. Garrick

NASA Grant NNX17AF08A Final Report

Executive Summary 3

Introduction 4

Formulation 5
Fluid dynamics
Particle fields
Nucleation
Coagulation
Condensation and evaporation
DBP vapor balance

Results 9
Flow configuration
Physical conditions
Numerical specifications
Physical Assumptions
Fluid flow, temperature, and mass fraction
Particle nucleation, number density, and size
Effect of particle settling
Effect of thermophoresis
Effect of heated vapor source orientation
Effect of flow speed
Smoke visualization

Summary 15

1 Executive Summary

This work examines how microgravity conditions on spacecraft change the development of smoke aerosol relative to smoke on Earth. We perform simulations of particle formation and growth with the goal of capturing smoke dynamics similar to those realized in experiments performed aboard the International Space Station. The simulations provide detailed information about fluid/thermal/particle interactions that cannot be acquired experimentally on orbit. A surrogate material, dibutyl pthalate, is expelled from a hot surface in a rectangular chamber while ambient air is blown through the domain, convecting the developing smoke downstream. The study considers two flow rates, two geometric configurations, and the isolated effects of gravitational particle settling and thermophoresis. In Earth gravity, buoyancy produces asymmetries in all fields of interest and amplifies vapor-to-particle conversion near room temperature surfaces at the upper boundary of the simulation domain. Gravitational settling removes large particles from the aerosol, limiting the maximum particle size to $\sim 40\ \mu\mathrm{m}$ and the mode particle size to $\sim 5\ \mu\mathrm{m}$. Without buoyancy or settling, the microgravity smoke plume is spatially symmetric and contains very large particles ($> 40\ \mu\mathrm{m}$). A bimodal particle size distribution results in microgravity when prominent recirculation occurs behind the hot vapor source. The distribution is comprised of large particles ($> 20\ \mu\mathrm{m}$) associated with long residency time, vapor rich, high temperature recirculation, and smaller particles ($< 10\ \mu\mathrm{m}$) that originate farther downstream of the hot vapor source. A distinct large particle mode does not appear in microgravity flows that do not have prominent high temperature recirculation, although the aerosol still contains large particles. Without recirculation, the mode particle diameter in microgravity is $\sim 10\ \mu\mathrm{m}$, or approximately twice that of terrestrial smoke. Results show that future experiments should carefully consider and quantify surface interactions and the presence of hot, recirculating vapor, in order to control geometry-dominated features of the particle size distribution.

A manuscript based on this work will be submitted to the journal Aerosol Science & Technology. The final document, when accepted, will be provided to NASA.

2 Introduction

Gravity affects the evolution of aerosols by introducing buoyancy and particle settling. Buoyant convection modifies the time-temperature history of the aerosol, as well as the spatial distribution of particles, condensing vapor, and thermal energy. Gravity also causes the settling of large particles out of the aerosol, provided the chemistry and vapor concentrations allow for the generation of large particles with appreciable settling velocities. These factors combine to increase particle residency times in high temperature, vapor rich regions in microgravity systems relative to terrestrial or Earth gravity systems (Urban et al., 1998, 2005). As a result, smoke particles produced in microgravity tend to be larger than those produced on Earth (Ku et al., 1995, Urban et al., 1998). Measurements suggest microgravity smoke particles to be $2-8$ times larger than terrestrial smoke particles (Urban et al., 2005). Moreover, large particles (not only smoke particles) persist in microgravity, with mass concentrations of particles with diameter $> 100\ \mu\mathrm{m}$ measured at ten-times that of a nominal terrestrial environment (Liu et al., 1991). Total particulate mass concentrations are also more than five times greater in microgravity (Liu et al., 1991). Understanding gravitational effects is of practical interest because it assists in the design of mission-critical smoke detection systems for use in microgravity environments, including the International Space Station (Mulholland et al., 2015); systems developed and calibrated for terrestrial use may not perform correctly in microgravity because of inappropriate particle size sensitivities (Urban et al., 2005).

The present work uses numerical simulation to explore the influence of gravity on pyrolysis smoke growth from a fundamental perspective. We simulate the formation and growth of dibutyl-phthalate (DBP) particles in Earth gravity and microgravity environments. We utilize direct numerical simulation via a Navier-Stokes nodal approach. The smoke originates from pyrolysis vapor expelled from a heated sample in a geometry similar to that considered in the SAME experiments (Mulholland et al., 2015). The intention of this effort is to quantify the differences between microgravity and Earth gravity smoke, and to identify the fluid dynamic mechanisms that cause the differences. This study considers two cross-flow velocities of sufficiently low speed that buoyancy cannot be neglected in the presence of gravity. Simulations are presented to assess the dependence of particle size distributions on thermophoresis, particle settling, and flow geometry. The paper is organized as follows: we introduce the simulation and modeling methodology, followed by a description of the flow geometry and the parameter space under consideration; we then examine differences in the wake geometry attributable to buoyancy for the lower of two flow speeds; next, we present and discuss instantaneous scalar fields as a means to describe particle accumulation and growth in the buoyant and non-buoyant wakes that form downstream of the hot vapor source; after developing an understanding of how the particle fields interact with the fluid dynamics for the low speed cases, we consider the effects of thermophoresis, particle settling, flow geometry, and the higher flow speed; we conclude with a discussion of the results, summarize findings relevant to future experimentation, and outline directions for future work.

3 Formulation

3.1 FLUID DYNAMICS

The gaseous systems are described by the compressible mass, momentum, and energy equations, considered here in dimensionless form. Conservation of mass is given by

$$\frac{\partial \rho^\star}{\partial t^\star} + \frac{\partial \rho^\star u_i^\star}{\partial x_i^\star} = 0, \tag{1}$$

where $\rho^\star$ is the density and $u_i^\star$ is the velocity (superscript asterisks denote dimensionless variables and subscripts denote vector components). Conservation of momentum is given by

$$\frac{\partial \rho^\star u_i^\star}{\partial t^\star} + \frac{\partial \rho^\star u_j^\star u_i^\star}{\partial x_j^\star} = -\frac{\partial P^\star}{\partial x_i^\star} + \frac{1}{Re}\frac{\partial \tau_{ij}^\star}{\partial x_j^\star} + \frac{1}{Fr^2}(\rho^\star - 1), \tag{2}$$

where $P^\star$ is the pressure, Re is the Reynolds number, $\tau_{ij}^\star$ is the viscous stress tensor, and Fr is the Froude number. The viscous stress tensor is defined by

$$\tau_{ij}^\star = \mu^\star \left(\frac{\partial u_j^\star}{\partial x_i^\star} + \frac{\partial u_i^\star}{\partial x_j^\star} \right) - \frac{2\mu^\star}{3}\frac{\partial u_k^\star}{\partial x_k^\star}\delta_{ij}, \tag{3}$$

where $\mu^\star$ is the dynamic viscosity and δ_{ij} is the Kronecker delta. The fluid is defined as an ideal gas with the corresponding equation of state,

$$P^\star = \rho_\alpha^\star R_\alpha^\star T^\star, \tag{4}$$

where $T^\star$ is the temperature and $R_\alpha^\star$ is the gas constant of species α (summation is implied by repeated subscript indicies). Conservation of total energy is given by

$$\frac{\partial \rho^\star E^\star}{\partial t^\star} + \frac{\partial \rho^\star u_j^\star H^\star}{\partial x_j^\star} = \frac{\mu^\star}{(\gamma - 1)M^2 RePr}\frac{\partial q_j^\star}{\partial x_j^\star} + \frac{1}{Re}\frac{\partial u_i^\star \tau_{ij}^\star}{\partial x_j^\star} + \frac{\partial}{\partial x_j^\star}\left(\frac{\rho^\star h_\alpha^\star}{ReSc}\frac{\partial Y_\alpha}{\partial x_j^\star} \right), \tag{5}$$

where $E^\star$ is the specific total energy, $H^\star$ is the specific total enthalpy, γ is the ratio of specific heats, M is the Mach number, Pr is the Prandtl number, $q_j^\star$ is the heat flux, Sc is the Schmidt number, Y_α is the mass fraction of species α, and $h_\alpha^\star$ is the specific enthalpy of species α. The heat flux is computed from Fourier's law,

$$q_j^\star = -k^\star \frac{\partial T^\star}{\partial x_j^\star}, \tag{6}$$

where $k^\star$ is the thermal conductivity and $T^\star$ is the temperature. The specific enthalpy is computed according to

$$h^\star = Y_\alpha \left(C_{p\alpha} T + h_\alpha^f \right) / h_o, \tag{7}$$

where h_α^f is the enthalpy of formation of species α, h_o is a reference enthalpy, and $C_{p\alpha}$ is the specific heat at constant pressure. The specific total energy is computed from

$$E^\star = \left(\frac{1}{2} u_i u_i + Y_\alpha \left(C_{v\alpha} T + e_\alpha^f \right) \right) / h_o, \tag{8}$$

where e_α^f is the energy of formation, and $C_{v\alpha}$ is the specific heat at constant volume. The specific total enthalpy $H^\star$ is analogously defined by

$$H^\star = \left(\frac{1}{2} u_i u_i + Y_\alpha \left(C_{p\alpha} T + h_\alpha^f \right) \right) / h_o, \tag{9}$$

where $C_{p\alpha}$ is the specific heat at constant pressure. Species transport is given by

$$\frac{\partial \rho^\star Y_\alpha}{\partial t^\star} + \frac{\partial \rho^\star Y_\alpha u_j^\star}{\partial x_j^\star} = \frac{\partial}{\partial x_j^\star}\left(\frac{\rho^\star}{ReSc}\frac{\partial Y_\alpha}{\partial x_j^\star} \right) + \dot{\omega}, \tag{10}$$

where $\dot{\omega}$ accounts for the conversion of species α vapor to the particle phase via nucleation and condensation.

3.2 PARTICLE FIELDS

Particles are transported in space by convection, diffusion, thermophoresis, and gravitational settling. Their size is affected by nucleation, coagulation, condensation, and evaporation. This work utilizes a nodal approximation to the general dynamic equation (Prakash et al., 2003), where the particle size distribution is discretized by a set of nodes k, each associated with a particle diameter d_k. The spatio-temporal transport of each particle size node k is considered with respect to its mass fraction $Y_k = \rho_k/\rho$,

$$\frac{\partial \rho^\star Y_k}{\partial t^\star} + \frac{\partial \rho^\star Y_k \left(u_j^\star + u_{j,k}^{tf^\star} + u_{j,k}^{ss^\star}\right)}{\partial x_j^\star} = \frac{\partial}{\partial x_j^\star}\left(\frac{\rho^\star}{ReSc_k}\frac{\partial Y_k}{\partial x_j^\star}\right) + \dot{\omega}_k, \tag{11}$$

where $u_{j,k}^{tf^\star}$ is the j component of the thermophoretic velocity of particle node k, $u_{j,k}^{ss^\star}$ is the j component of the Stokes settling velocity of particle node k, Sc_k is the particle Schmidt number, and $\dot{\omega}_k$ is a source term that accounts for transport in size space (this term will be defined in Section 2.6). The particle diffusivity is defined by

$$\mathcal{D}_k = \frac{K_b T C_k}{3\pi\mu d_k}, \tag{12}$$

where K_b is Boltzmann's constant, C_k is the Cunningham correction factor, and d_k is the diameter of particles in node k. The Cunningham correction factor is taken as

$$C_k = 1 + Kn_k\left(1.142 + 0.558\exp\left(-0.999/Kn_k\right)\right) \tag{13}$$

where Kn_k is the Knudsen number of particles in node k

$$Kn_k = 2\lambda/d_k. \tag{14}$$

The mean free path in air λ is computed from

$$\lambda = \lambda_o\left(\frac{P_{atm}}{P}\right)\left(\frac{T}{T_1}\right)^2\left(\frac{T_1 + T_s}{T + T_s}\right) \tag{15}$$

where P_{atm} is the atmospheric pressure and the model parameters are $T_s = 110.4K$, $T_1 = 296.15K$, and $\lambda_o = 67.3nm$ (Otto et al., 1999).

The Talbot equation defines the thermophoretic velocity,

$$u_{j,k}^{tf^\star} = -k^{tf}\frac{\mu}{\rho U_o T}\frac{\partial T}{\partial x_j}, \tag{16}$$

where

$$k^{tf} = \frac{2.294C_k\left(k/k_p + 2.2Kn_k\right)}{\left(1 + 3.493Kn_k\right)\left(1 + 2k/k_p + 4.4Kn_k\right)}, \tag{17}$$

and k_p is the thermal conductivity of the particulate phase (Pyykönen and Jokiniemi, 2000, Talbot et al., 1980). The dimensionless Stokes settling velocity, $u_{j,k}^{ss^\star}$, is defined by

$$u_{j,k}^{ss^\star} = \frac{\left(\rho_c - \rho\right)g_j d_k^2}{18\mu U_o}, \tag{18}$$

where g_j is the gravitational acceleration in the j direction.

The source term $\dot{\omega}_k$ controls transport in size space due to nucleation, coagulation, condensation, and evaporation,

$$\dot{\omega}_k = \left(\omega_k^N + \omega_k^c + \omega_k^s\right) \times \frac{L_o m_k}{U_o \rho_o}, \tag{19}$$

where m_k is the mass of a particle in node k, ω^N is the number density change due to nucleation, ω^c is the number density change due to coagulation, and ω^s is the number density change due to the combined effects of condensation and evaporation.

3.3 NUCLEATION

We consider the classical nucleation expression modified by a self-consistent correction proposed by Girshick and Chiu (1990). The nucleation rate is scaled by a material-specific enhancement factor, c_{nuc}, resulting in a nucleation rate J defined by

$$J = c_{nuc}\frac{(\rho Y)^2}{\rho_c m_m}\sqrt{\frac{2\sigma}{\pi m_m}}\frac{1}{S}\exp\left(\frac{\sqrt[3]{36\pi v_m^2}\sigma}{k_B T} - \frac{\pi\sigma d_c^2}{3k_B T}\right), \tag{20}$$

where d_c is the critical diameter of nucleating particles given by

$$d_c = \frac{4\sigma v_m}{k_B T \ln(S)}. \tag{21}$$

The nucleation rate depends on the mass m_m and volume v_m of a vapor molecule, the surface tension σ, and the saturation ratio S. Particles nucleating with size d_c do not correspond exactly to any of the discretized size nodes. Following Prakash et al. (2003), the nucleated particles are placed in the node k with a diameter d_k that is minimally larger than d_c. This procedure results in the nucleation source term for node k,

$$\omega_k^N = \frac{d_c^3}{d_k^3} J. \tag{22}$$

The nucleation source term for all other nodes is zero.

3.4 COAGULATION

The rate of change of particles in size class k due to coagulation is defined by

$$\omega_c^k = \frac{1}{2}\sum_{i=2}^{N_s}\sum_{j=2}^{N_s}\chi_{ijk}\beta_{ij}Q_i Q_j - Q_k\sum_{i=2}^{N_s}\beta_{ik}Q_i \tag{23}$$

where Q_i is the number density of node i, β_{ij} is the collision frequency function corresponding to nodes i and j, and χ_{ijk} is the mass splitting function defined by (Prakash et al., 2003)

$$\chi_{ijk} = \begin{cases} \frac{v_{k+1}-(v_i+v_j)}{v_{k+1}-v_k} & \text{if } v_k \le v_i + v_j < v_{k+1} \\ \frac{(v_i+v_j)-v_{k-1}}{v_k - v_{k-1}} & \text{if } v_{k-1} \le v_i + v_j < v_k \\ 0 & \text{else.} \end{cases} \tag{24}$$

The Dahneke coefficients are used to extend the near-continuum frequency function β_{co} to all relevant values of Kn (Otto et al., 1999). The collision coefficients are given by

$$\beta_{ij} = \beta_{co}\frac{1 + Kn_D}{1 + 2Kn_D + 2Kn_D^2}, \tag{25}$$

where β_{co} is the near-continuum collision frequency function,

$$\beta_{co} = \frac{2k_B T}{3\mu}\left(v_i^{1/3} + v_j^{1/3}\right)\left(\frac{C_i}{v_i^{1/3}} + \frac{C_j}{v_j^{1/3}}\right). \tag{26}$$

The enhancement Knudesen number Kn_D is defined as

$$Kn_D = \frac{4\mathcal{D}_{ij}}{\bar{c}_{ij} d_{ij}}, \tag{27}$$

where $\mathcal{D}_{ij} = \mathcal{D}_i + \mathcal{D}_j$ and $d_{ij} = d_i + d_j$ and $\bar{c}_{ij} = \sqrt{\bar{c}_i^2 + \bar{c}_j^2}$. The mean molecular speed $\bar{c}$ is given by $\bar{c}_k = \sqrt{8k_B T/\pi m_k}$.

3.5 CONDENSATION AND EVAPORATION

The saturation monomer number density with respect to particles of size d_k is computed according to

$$Q_{1,k}^s = \frac{P_s}{k_b T} \exp\left(\frac{4\sigma \mathrm{MW}}{RT\rho_c d_k}\right), \tag{28}$$

where P_s is the saturation pressure and MW is the molecular weight of the condensing species. Using this saturation concentration, the rate of change of particles in size class k due to condensation and evaporation is given by (Prakash et al., 2003)

$$\omega_s^k = \begin{cases} \frac{v_m}{v_k - v_{k-1}} \beta_{1,k-1} \left(Q_1 - Q_{1,k-1}^s\right) Q_{k-1} & \text{if } Q_1 > Q_{1,k-1}^s \\ -\frac{v_m}{v_{k+1} - v_k} \beta_{1,k+1} \left(Q_1 - Q_{1,k+1}^s\right) Q_{k+1} & \text{if } Q_1 < Q_{1,k+1}^s \\ -\frac{v_m}{v_{k+1} - v_k} \beta_{1,k} \left(Q_1 - Q_{1,k}^s\right) Q_k & \text{if } Q_1 > Q_{1,k}^s \\ \frac{v_m}{v_k - v_{k-1}} \beta_{1,k} \left(Q_1 - Q_{1,k}^s\right) Q_k & \text{if } Q_1 < Q_{1,k}^s \end{cases}, \tag{29}$$

where Q_1 is the number density of monomers and Q_k is the number density of particles of size class k.

3.6 DBP VAPOR BALANCE

Nucleation, condensation, and evaporation are interactions between the vapor and particle phases. The source term in the vapor transport equation, Eqn. (10), is given by

$$\dot{\omega} = -\left(\frac{d_c^3}{d_m^3} J + \sum_{k=2}^{N} \frac{v_k}{v_m} \omega_s^k\right) \times \frac{L_o m_m}{U_o \rho_o} \tag{30}$$

where the first term on the right-hand-side is the sink due to nucleation and the second term on the right-hand-side is the source associated with condensation and evaporation (Prakash et al., 2003).

4 Results

4.1 FLOW CONFIGURATION

We simulate the formation and growth of DBP particles in air flowing over a heated sample that expels DBP vapor. The simulation mimics the smoldering or pre-conflagration stage of a fire studied in experiments performed aboard the ISS (Mulholland et al., 2015). A schematic of the computational domain is shown in Fig. 1. Room temperature air enters a long chamber and flows over a heated rectangular sample. Vapor released from the sample nucleates into particles that grow and are subsequently collected downstream for analysis. The flow is laminar and steady, allowing the two-dimensional representation to present physical characteristics largely representative of the SAME experiments (Mulholland et al., 2015). DBP is used as the surrogate material because its properties are known and its use in modeling is well-established (unlike the materials used in the SAME experiments).

4.2 PHYSICAL CONDITIONS

Constant fluid properties for air and DBP are provided in Table 1, and variable DBP properties are provided in Table 2. The enhancement factor required for the nucleation expression, Eqn. (20), is defined for DBP as $c_{nuc} = 3.2 \times 10^{-4}$ (Nguyen et al., 1987, Pyykönen and Jokiniemi, 2000).

The thickness of the heated DBP source serves as the reference length scale, $L_o = 0.0044\ \mathrm{m}$ (illustrated in Fig. 1), which is representative of the length scale in the SAME experiments (Mulholland et al., 2015). The air enters from the left with velocity U_o and temperature $T_o = 300\ \mathrm{K}$. We consider two inlet velocities: $U_o = 0.08\ \mathrm{m/s}$, the velocity considered in the SAME experiments, and a slightly faster flow of $U_o = 0.10\ \mathrm{m/s}$. The upper and lower boundaries are impermeable, no-slip, iso-thermal surfaces ($T_{\mathrm{wall}} = T_o$). The heated rectangle is also no-slip, has a fixed temperature of $T_s = 473.2\ \mathrm{K}$, and releases DBP vapor at a constant mass flux of $q'' = 8.4 \times 10^{-5}\ \mathrm{kg/m^2 \cdot s}$. The mass flux has been selected to be representative of the mass loss data presented by Mulholland et al. (2015). Zero-derivative conditions are imposed for all species and particle fields at the inlet and outlet. The gravitational acceleration is considered at terrestrial ($g_o = 9.81\ \mathrm{m/s^2}$) and microgravity ($g_o = 0\ \mathrm{m/s^2}$) conditions.

The inlet velocities and the gravitational accelerations considered represent four cases with varying Reynolds and Froude numbers. A summary of the simulations performed at the two flow speeds and gravitational levels is given in Table 3. The flags "EG" and "MG" identify the "Earth gravity" and "microgravity" conditions, while 1 and 2 identify the $U_o = 0.08\ \mathrm{m/s}$ and $U_o = 0.1\ \mathrm{m/s}$ conditions, respectively. Case 1MG is therefore defined to have an inlet velocity of $U_o = 0.08\ \mathrm{m/s}$ and $g_o = 0\ \mathrm{m/s^2}$, corresponding to a Reynolds number $Re = 22.3$ and a Froude number $Fr = \infty$. In addition to the four base cases listed in Table 3, we examine the aerosol development sensitivity to: (1) particle settling, (2) thermophoresis, and (3) the flow geometry. For this purpose, a series of additional cases, defined in Table 3, are constructed by modifying the setup of Case 1MG and/or Case 1EG. Case 1EG-NS (where NS stands for "no settling") is identical to Case 1EG, but the settling mechanism has been neglected. Case 1MG-NT (where NT stands for "no thermophoresis") is identical to Case 1MG, but the thermophoresis transport mechanism has been neglected. Cases 1MG-R and 1EG-R (where R stands for "rotated") are identical to Cases 1MG and 1EG, except the heated vapor source has been rotated 90°. In all cases, data is collected and presented at a time of $t = 7$ seconds.

4.3 NUMERICAL SPECIFICATIONS

The domain is non-uniformly discretized by 960 grid points in the x direction and 240 grid point in the y direction. The governing equations are solved on this computational mesh with a second-order MacCormack scheme. The particle size space is discretized by 45 particle nodes, with the volume of each successive node increased by a factor of 2.25, i.e. $v_k = 2.25 \times v_{k-1}$. The volume of the first node is selected to be the volume of a DBP molecule, $v_1 = v_m = 4.44 \times 10^{-28}\ \mathrm{m^3}$. Boundary conditions are

implemented by considering the characteristic relations for the Euler equations accounting for gravity (Poinsot and Lele, 1992). The computational tool used for this work has been well-validated and used for prior studies of aerosols (Garrick, 2011, 2015, Garrick and Khakpour, 2004, Garrick et al., 2001, Khakpour and Garrick, 2003, Modem and Garrick, 2003).

4.4 PHYSICAL ASSUMPTIONS

All assumptions utilized in this work are listed below.

- From the perspective of thermal conductivity and viscosity, the condensing vapor phase is considered sufficiently dilute such that μ and k are computed for air as a function of temperature and density (Kadoya et al., 1985). Both the air and the condensing vapor are considered calorically perfect gases with constant properties defined in Table 1.
- The Mach number M is only important for flows with $M \gtrsim 0.3$. Although the physics are insensitive to variation in Mach number when the Mach number is small, the ratio of sound speed to convective speed imposes a large computational burden. The computational challenge is often remedied with a Boussinesq approximation, but this approach is not valid in the presence of large density variations. Additionally, an elliptical solve for pressure is computationally intractable for the number of iterations required to resolve the particle time scales. We instead use a scaled dynamic pressure as the reference pressure $P_o = \rho_o \left(c_1 U_o\right)^2$, such that $P^\star = \frac{\rho_\alpha R_\alpha T}{P_o}$, where c_1 is specified to reduce the effective speed of sound to $8\ \mathrm{m/s}$. The flow field was found to be independent of c_1 for the values used.
- All particles are assumed to be spherical.
- Condensed phase DBP properties are defined by their continuum values.
- There is no momentum exchange between the particles and the fluid (the air/DBP vapor mixture); the particle velocity equals the fluid velocity, augmented by the thermophoretic velocity and the stokes settling velocity. The stokes settling velocity is assumed to apply for all particles in the domain (including the large particles).
- The fluid and particles are in thermal equilibrium and have the same temperature.
- Particle diffusion coefficients decrease rapidly with diameter. This poses a numerical challenge due to the lack of a physical mechanism for diffusing numerical errors. For this reason, the minimum diffusion coefficient considered in the particle mass fraction transport equations corresponds to the fifth particle size node – any node greater than the fifth node is assigned the particle diffusion coefficient of the fifth node.
- The mass flux of DPB from the heated rectangle is spatially uniform and all surfaces have spatially uniform temperature.
- Particles that impact a solid surface "stick" with 100% efficiency, and are removed from the flow.
- The aerosol does not achieve a rigorous steady-state because particle evolution continues (slowly) in regions of the flow with recirculation, particularly behind the hot vapor source. The recirculation is characterized by high temperatures, long particle residency times, and the continuous introduction of additional vapor. The hydrodynamic and thermal fields, conversely, become steady after only a few seconds. Although a rigorous steady-state does not exist, all aerosol features develop and effectively stabilize shortly after the hydrodynamic fields. We therefore present all results at time $t = 7$ seconds, which is considered to be the effective steady-state. At his time, all fields are either steady, or changing at a rate that does not affect the conclusions drawn from the simulations.

4.5 FLUID FLOW, TEMPERATURE, AND MASS FRACTION

The velocity field controls the convection of thermal energy, condensible species vapor, and particles. Vorticity ω, defined as the curl of the velocity vector

$$\omega = \nabla \times \mathbf{u}, \tag{31}$$

is positive in regions of the flow where fluid rotates counterclockwise, and is negative in regions where the fluid rotates clockwise. The sign and magnitude of vorticity therefore identifies the location, orientation, and intensity of vortices and boundary layers.

Figure 2 shows the steady dimensionless vorticity. Panel (a) shows Case 1MG and panel (b) shows Case 1EG. The grey rectangles represent the heated vapor source. Red fluid rotates counterclockwise (positive vorticity) and blue fluid rotates clockwise (negative vorticity). Immediately upon entering the channel, the fluid at the walls is brought to rest by the no slip condition, signified by the large vorticity gradients in the upper and lower left corners of panels (a) and (b). As the fluid travels downstream from left to right, boundary layers develop symmetrically on the upper and lower walls due to viscous diffusion. Further downstream, the heated rectangle reduces the cross-sectional flow area, and the pressure field accelerates the fluid through flow paths above and below the heated vapor source. The microgravity case in panel (a) remains symmetric near the heated vapor source, whereas the Earth gravity case in panel (b) develops an asymmetry with the majority of fluid passing below the heated vapor source (indicated by larger vorticity magnitudes below than above). Comparatively low vorticity values on the downstream side of the rectangles indicate the presence of low speed, recirculating wakes. The microgravity wake in panel (a) is symmetric, whereas the Earth gravity wake in panel (b) is asymmetric. As the fluid continues to flow downstream, the boundary layers thicken in both cases and the fluid exits the domain to the right. Variations in density due to heat transfer from the rectangle have influenced the vorticity fields via baroclinic torque and buoyancy. Because there is no gravity in Case 1MG in panel (a), density variations due to heating do not disturb the symmetry of the flow field. The asymmetry apparent in panel (b) is due to buoyancy effects.

Temperature fields with overlaid velocity vectors are shown in Fig. 3 at time $t = 7$ seconds. Panel (a) shows Case 1MG and panel (b) shows Case 1EG. The downstream portion of the domain has been expanded for clarity. Velocity vectors in panel (a) show a symmetric wake profile with two counter-rotating vortices surrounded by dividing streamlines. Panel (b) shows an asymmetric buoyant wake, also with two counter rotating vortices: one near the lower right corner of the heated vapor source, and another near the upper right corner of the heated vapor source. Both panels show hot fluid in white, warm fluid in red, and room temperature fluid in black. Thermal energy diffuses from the heated vapor source to the adjacent fluid. Hot fluid convects energy downstream and cools while diffusing energy to cooler fluid and the isothermal upper and lower walls. The highest temperatures appear near the heated surface and in the wake. In both cases, the gas in the center of the channel exits the domain at greater than 350 Kelvin. The microgravity case maintains a symmetric temperature field because there is no gravity, and therefore no asymmetric coupling between the vorticity and temperature via buoyancy. In panel (b), buoyant convection resulting from gravitational forces caries the low density, hot fluid towards the upper surface. As the hot fluid encounters the isothermal upper surface, a thin thermal boundary layer develops. Hot fluid cools rapidly when in close proximity to the upper surface due to the thin thermal boundary layer.

DBP vapor mass fraction is shown in Fig. 4 at time $t = 7$ seconds. Panel (a) shows Case 1MG and panel (b) shows Case 1EG. Consistent with the vorticity and temperature, the DBP vapor mass fraction is symmetric in Case 1MG and asymmetric in Case 1EG. The highest mass fractions of DBP (approximately 0.05) appear in white on the immediate downstream side of the heated vapor source in both panels (a) and (b). The recirculating wakes in both cases contain a small concentration of DBP vapor, but are encompassed by streaks of DBP vapor issuing from the upper and lower surfaces of the heated vapor source. In Case 1MG, the DBP vapor streaks interact via diffusion at the downstream side of the wake and eventually merge. In Case 1EG, merging velocity characteristics, indicated by the merging velocity vectors at the downstream side of the wake, rapidly combine the upper and lower vapor streaks. In both Case 1MG and Case 1EG, the majority of the vapor is consumed before the domain outlet.

4.6 PARTICLE NUCLEATION, NUMBER DENSITY, AND SIZE

The particle nucleation rate is a function of vapor mass fraction and temperature (and pressure). Figure 5 shows the DBP nucleation rate at time $t = 7$ seconds for Case 1MG in panel (a) and Case 1EG in panel (b). Panel (a) shows that nucleation occurs exclusively downstream of the heated vapor source in Case 1MG, and most rapidly at a rate of $\sim 10^{12}\ \left[\mathrm{m}^3 \cdot \mathrm{s}\right]^{-1}$ near the periphery of the recirculating wake. Nucleation does not occur at a significant rate anywhere in the wake. The nucleation rate decreases quickly after approximately ten dimensionless length units downstream of the heated vapor source. Panel (b) shows two primary zones of nucleation in Case 1EG: downstream of the upper wake vortex and in the region downstream and above the upper wake vortex, near the upper surface. Nucleation occurs at a higher rate, $\sim 10^{14}\ \left[\mathrm{m}^3 \cdot \mathrm{s}\right]^{-1}$, in Case 1EG than in Case 1MG.

Particle number densities at time $t = 7$ seconds are shown in Fig. 6 for Cases 1EG and 1MG in panels (a) and (b), respectively. Panel (a) shows a particle number density of $\sim 10^{10}\ \mathrm{m}^{-3}$ at the center of each counter-rotating wake vortex. Thermophoretic forces prevent particles from contacting the downstream surface of the heated vapor source (this will be demonstrated in Section 4.8), resulting in a void of particles between the heated vapor source and the particle-dense wake vortices. Particles are also less numerous on the downstream side of the recirculating wake: particles do not nucleate in this region, dividing streamlines prevent particles from being convected into this region, and low diffusion rates limit particles from rapidly diffusing into this region. Even further downstream, the particle number density remains symmetric, with the largest particle number densities of $\sim 10^{11}\ \mathrm{m}^{-3}$ appearing slightly separated from the symmetry line on top and bottom. No particles appear near the upper and lower surfaces. Panel (b) shows the particle number densities for Case 1EG. More particles accumulate in the upper wake vortex than in the lower wake vortex. The Case 1EG wake contains fewer particles than the Case 1MG wake. In Case 1EG, a high particle concentration of nearly $\sim 10^{12}\ \mathrm{m}^{-3}$ appears over a large area near the upper surface due to the high nucleation rate in the region. Significantly more particles appear in this near-wall region than anywhere in the Case 1MG particle field. Unlike Case 1MG, Case 1EG contains particles throughout the entire vertical span of the flow chamber: particles that appear in lower concentrations near the bottom wall have arrived there via the gravitational settling mechanism. Particles that settle are sufficiently large to be removed from the aerosol via gravity. This can be demonstrated by considering the spatial distribution of particle size.

The diameter of average mass d_m is defined by

$$d_m = \left(\frac{6}{\pi} \frac{\sum_{i=2}^{N_s} v_i Q_i}{\sum_{i=2}^{N_s} Q_i} \right)^{1/3}. \tag{32}$$

Figure 7 shows the diameter of average mass at time $t = 7$ seconds for Case 1MG in panel (a) and Case 1EG in panel (b). The diameter of average mass is typically $10 - 20\ \mu\mathrm{m}$ in both cases. The diameter of average mass is notably smaller in regions where nucleation rates or particle number densities are large (near the upper surface in Case 1EG, for example). The Case 1EG wake contains only isolated regions of large particles, including in the upper and lower vorticies, while the entirety of the Case 1MG wake contains large particles. A signification region downstream and below the heated vapor source contains very large particles in Case 1EG: particles sufficiently large to rapidly settle out of the flow dominate this region. Downstream of the Case 1MG wake, the diameter of average mass depends significantly on cross-stream position, but not on stream-wise position. The largest particles appear on the symmetry line, corresponding to a region of low particle number density.

The particle size distribution (PSD) provides a quantitative measure of particle number and size. We construct the size distribution by numerically "collecting" a volume of the aerosol and counting the number of particles in each node k contained in the sample. The sample is collected by computing the flux of particles through a vertical plane located at x', where x' is the dimensionless distance from the downstream surface of the heated element, indicated in Fig. 1. The collected number density $\bar{Q}_k\ \left[m^{-3}\right]$ of particles belonging to size k is therefore

$$\bar{Q}_k = \frac{1}{h} \int_{-h/2}^{h/2} u\,(y)\, Q_k\,(y)\, dy, \tag{33}$$

where $h = 6.82 L_o$ is the height of the domain and u is the x-component of the velocity vector. The collected number density $\bar{Q}_k$ is therefore the number of particles collected per unit time and per unit volume at the measurement plane x'.

Figure 8 shows the particle number densities for Cases 1MG and 1EG at downstream measurement plane $x' = 26$ at time $t = 7$ seconds. Case 1MG is shown in red and Case 1EG is shown in black. The Case 1MG distribution contains particle diameters ranging from hundreds of nanometers to tens of micrometers, with a small percentage of particles approaching one hundred micrometers. The most frequent size is approximately $5\ \mu\mathrm{m}$. The Case 1MG distribution contains two local maximums: one at $5\ \mu\mathrm{m}$ and another at $30\ \mu\mathrm{m}$. The Case 1EG distribution contains only one local maximum at $5\ \mu\mathrm{m}$. Case 1EG contains a larger number density than Case 1MG for all particle sizes $< 20\ \mu\mathrm{m}$, and Case 1EG contains no particles $> 40\ \mu\mathrm{m}$ – these large particle settle out of the system. Case 1EG contains orders of magnitude more particles $< 5\ \mu\mathrm{m}$ than Case 1MG.

4.7 EFFECT OF PARTICLE SETTLING

In order to isolate the effects of particle settling, we modify Case 1EG by neglecting the settling mechanism (by setting $u_{j,k}^{ss^\star} = 0$). This case is referred to as Case 1EG–NS, where NS stands for "no settling". Figure 9 compares size distributions for Case 1MG, Case 1EG, and Case 1EG–NS at measurement plane $x' = 26$ at time $t = 7$ seconds. The Case 1EG–NS distribution is considerably wider than the other cases. It contains particles ranging in size from nanometers to $100\ \mu\mathrm{m}$. Case 1EG–NS also contains two local maxima in the particle size distribution, much like Case 1MG and unlike Case 1EG. Eliminating the settling mechanism has expanded the size distribution to include both smaller and larger particles, and has also introduced a new local maximum in the size distribution near $35\ \mu\mathrm{m}$.

4.8 EFFECT OF THERMOPHORESIS

Figure 10 shows particle number densities at time $t = 7$ seconds for Case 1MG in panel (a) and Case 1MG-NT in panel (b) (where NT stands for "no thermophoresis"). When thermophoresis is active, as in panel (a), particles do not appear adjacent to the surface of the heated vapor source; the temperature gradient is large normal to the surface, and thermophoresis forces the particles out of this region. When thermophoresis is inactive, particles are able to remain near the hot surface. In this region, the temperature is sufficiently high to cause evaporation. Small particles are able to diffuse out of the wake, resulting in lower particle number densities inside the Case 1MG-NT wake than the Case 1MG wake. Downstream of the wake, larger particle number densities are present in Case 1MG-NT relative to Case 1MG, following from the decreased number of particles trapped in the Case 1MG-NT wake vorticies.

Figure 11 shows particle size distributions for Cases 1MG and 1MG-NT at time $t = 7$ seconds. Consistent with Fig. 10, the peak number density in Case 1MG-NT is greater than Case 1MG by approximately a factor of five – the most frequent particle size is $\sim 5\ \mu\mathrm{m}$ for both cases. A local maximum appears in both distributions around $\mathrm{d_p} = 30\ \mu\mathrm{m}$. This local maximum contains fewer particles in Case 1MG-NT than in Case 1MG, potentially because this local maximum is comprised of particles that have interacted with the wake region. In Case 1MG-NT, some of these particles have undergone evaporation and shifted to smaller particle sizes.

4.9 EFFECT OF HEATED VAPOR SOURCE ORIENTATION

Figure 12 shows the particle number densities at time $t = 7$ seconds for Case 1MG-R in panel (a) and Case 1EG-R in panel (b) (where R stands for "Rotated"). Cases 1MG-R and 1EG-R are identical to Cases 1MG and 1EG, except the heated vapor source has been rotated $90°$. The wake regions of Cases 1MG-R and 1EG-R are considerably smaller than Cases 1MG and 1EG, and neither Case 1MG-R nor Case 1EG-R contain particles in their wakes. The results from Case 1MG-R differ from Case 1MG in that they do not prominently display two high particle number density streaks around the symmetry line, but instead contain large particle number densities everywhere near the symmetry plane. Case 1EG-R presents large particle number densities near the cool upper surface, similar to Case 1EG. Unlike Case 1EG, Case 1EG-R does not contain particles inside its comparatively small downstream wake region, but instead contains particles in a large recirculation region that appears above the heated vapor source. The number density of particles in the recirculation region directly above the heated vapor source is lower than the particle number density downstream of the recirculation region. Settling

particles are not apparent, as in Case 1EG, which may be attributable to the lack of a wake region.

Figure 13 shows particle size distributions for Cases 1MG-R and 1EG-R at time $t = 7$ seconds. Both distributions present a single local maximum in particle size – the local maximum prominent at $d_p = 30\ \mu\mathrm{m}$ for Case 1MG is not present in Case 1MG-R. Consistent with Case 1EG, Case 1EG-R contains fewer particles larger than approximately $d_p = 10\ \mu\mathrm{m}$, and no particles larger than $d_p = 40\ \mu\mathrm{m}$. The most frequent particle size for Case 1EG-R is the same as Case 1EG ($d_p = 5\ \mu\mathrm{m}$). The most frequent particle diameter for Case 1MG-R is $d_p = 10\ \mu\mathrm{m}$ – approximately two times larger than Cases 1MG, 1EG, and 1EG-R.

4.10 EFFECT OF FLOW SPEED

The Case 1 simulations were performed with an $8\ \mathrm{cm/s}$ inlet flow velocity. Case 2MG and Case 2EG consider a $10\ \mathrm{cm/s}$ inlet flow velocity. Particle number densities and velocity vectors for Case 2MG and Case 2EG are shown in Fig. 14 at time $t = 7$ seconds. Panel (a) shows Case 2MG and panel (b) shows Case 2EG. Case 2MG presents many features similar to Case 1MG, including symmetry about the center line and high particle number densities in the wake vortices. Case 2MG has an elongated wake associated with a higher flow speed. Case 2EG differs from Case 1EG in that particles are not prominent near the upper surface until much farther downstream – this is attributable to the decrease in Froude number, and corresponding decrease in upward motion due to buoyancy. Although the particle field remains asymmetric, the increased flow velocity has decreased the influence of buoyancy, reducing interaction with the upper surface. The presence of particles in the lower portion of the domain, as in Case 1EG, is attributable to the gravitational settling dynamic.

Particle size distributions for Case 2MG and Case 2EG computed at $x' = 26$ and time $t = 7$ seconds are shown in Fig. 15. The most common particle size is $\sim 5\ \mu\mathrm{m}$ for both cases. Case 2EG, unlike Case 1EG, contains particles on the nanometer scale – nucleation begins near the upper surface further downstream, and particles resulting from nucleation remain in the flow near the measurement plane $x' = 26$.

4.11 SMOKE VISUALIZATION

The simulated smoke is visualized by computing the particle surface area density $\overline{S}(\mathbf{x})$ according to

$$\overline{S}(\mathbf{x}) = \sum_{k=2}^{N_s} \pi d_k^2 Q_k(\mathbf{x}). \tag{34}$$

The particle surface area density $\overline{S}(\mathbf{x})$ is shown in Fig. 16 at time $t = 7$ seconds. Panel (a) shows Case 1MG, panel (b) shows Case 1EG, panel (c) shows Case 2MG, and panel (d) shows Case 2EG. Darker regions correspond to large smoke concentrations and white areas contain no smoke. High concentrations of smoke are apparent in the wake vortices of all cases. The microgravity cases in panels (a) and (c) contain larger and darker wake regions than the Earth gravity cases in panels (b) and (d). Panel (b), corresponding to Case 1EG, presents high smoke concentrations near the upper wall. The visible smoke in Case 1EG also occupies a larger downstream area than the other cases, resulting from the high nucleation rates near the upper surface. When gravitational forces dominate, wall effects can be significant, as demonstrated by case Case 1EG in panel (b). In Earth gravity, the smoke is more prominent downstream than in the wake region. Visual inspection also suggests the Earth gravity smoke to be more sensitive to small perturbations in flow speed than the microgravity smoke (in the configuration tested).

5 Summary

We have simulated smoke formation and growth in microgravity and Earth gravity to examine how buoyancy and particle settling affect thermal/fluid/particle interactions and the resultant smoke particle size distributions. The simulated geometry emulates the SAME experiments performed aboard the International Space Station (Mulholland et al., 2015). Dibutyl pthalate (DBP) vapor is released from a heated rectangular sample, which serves as a surrogate for the smoldering material used in the SAME experiments.

Large particles ($> 30\ \mu\mathrm{m}$) are present in the wake region downstream of the heated vapor source, and large particle number densities ($\sim 10^{10}\ \mathrm{m}^{-3}$) occur inside the wake vortices. The wake contains large particle number densities because dividing streamlines prevent the particles from being convected away, and the particles continue to grow as condensation consumes all available DBP vapor. The particles simultaneously grow via coagulation. Downstream of the wake, two parallel particle streams contain high number densities of both large and small particles. No particles are present far downstream with diameter $< 400\ \mathrm{nm}$ because nucleation only occurs for a short distance downstream of the heated vapor source. The particle size distributions contain local maximums near $\mathrm{d_p} = 5\ \mu\mathrm{m}$ and $\mathrm{d_p} = 35\ \mu\mathrm{m}$ for both flow speeds. The faster flow conditions present a slightly larger wake but this does not manifest as a meaningful difference in particle size distribution. Thermophoresis prevents particles from contacting the heated surface, which has been demonstrated by turning off the thermophoresis mechanism. Without thermophoresis, particles contact the heated vapor source and undergo evaporation. With respect to the particle size distribution, this has the effect of decreasing the number of large particles and increasing the number of small particles – this supports an assertion that the wake is responsible for producing the largest particles in the system.

When gravity is present, all fields become asymmetric due to buoyant convection. Upward convection in the faster flow brings DBP vapor into close proximity with the cool upper surface, resulting in locally high nucleation rates near the upper wake vortex and the upper surface. Large particle number densities ($\sim 10^{11}\ \mathrm{m}^{-3}$) are present along the upper surface, and slightly lower number densities are present in the upper wake vortex. Comparably few particles are present in the lower wake vortex, where the diameter of average mass is maximized. Downstream of the wake vortices, gravitational settling eliminates all particles with diameter $> 35\ \mu\mathrm{m}$. Unlike the microgravity case, the $8\ \mathrm{cm/s}$ Earth gravity case is not bimodal, with a single local maximum at $\mathrm{d_p} = 5\ \mu\mathrm{m}$. Without particle settling, a bimodal distribution results with two local maximums at particle diameters comparable to the microgravity case. Rotating the heated vapor source 90° dramatically reduces the wake volume, and no particles appear in the small wake. Rotating the heated vapor source does not affect the Earth gravity case as significantly as the microgravity case. Increasing the flow speed from decreases the prominence of buoyancy, delays interaction with the upper surface until further downstream, widens the particle size distribution to include particles on the nanometer scale, and introduces a second local maximum to the PSD near $\mathrm{d_p} = 30\ \mu\mathrm{m}$.

Microgravity smoke presented in this study contains larger particles than terrestrial smoke. When the microgravity PSD is bimodal, the larger mode contains particles with $\mathrm{d_p} > 35\ \mu\mathrm{m}$, while the terrestrial PSD contains effectively zero particles in this size range. Alternatively, the microgravity PSD can be unimodal with a mode diameter of $\mathrm{d_p} = 10\ \mu\mathrm{m}$ – roughly twice that of the terrestrial case. Results exhibit three primary mechanisms that differentiate the microgravity and Earth gravity smoke: (1) different recirculation geometries and volumes modify the time-temperature-concentration history of the aerosol in regions of high temperature and high vapor mass fraction; (2) particle settling eliminates large particles from Earth gravity smoke, which reduces the maximum particle size and modifies aerosol development by changing coagulation and condensation rates of the remaining particles; (3) buoyant convection can transport condensible species toward relatively cool surfaces, increasing vapor-to-particle conversion. Each of these factors tend to increase the size of particles in microgravity relative to Earth gravity. However, each of these factors is also highly material- and geometry-dependent: (1) the wake dynamic becomes unimportant if nucleation occurs far downstream of the wake, or if there is a time dependent flow without a stable recirculation region; (2) particle settling is irrelevant if the condensing species forms particles that are too small to settle (the DBP considered in this work forms large particles); (3) if smoke is produced in an open environment, there may not be cool surfaces nearby to increase nucleation rates. In these situations, microgravity and Earth gravity smoke are hypothesized to be more similar. In

contrast, a quiescent, enclosed system is a relevant case where Earth gravity and microgravity smoke are hypothesized to differ significantly, and is a consideration of future work.

References

S. C. Garrick. Effects of turbulent fluctuations on nanoparticle coagulation in shear flows. *Aerosol Sci. and Technol.*, 45:1271–1285, 2011. doi: $10.1080/02786826.2011.589482$.

S. C. Garrick. Growth mechanisms of nano-structured titania in turbulent reacting flows. *J. Nanotechnol.*, 2015:1–10, 2015. doi: $10.1155/2015/642014$.

S. C. Garrick and M. Khakpour. The effects of differential diffusion on nanoparticle coagulation in temporal mixing layers. *Aerosol Sci. Technol.*, 38(8):851–860, August 2004.

S. C. Garrick, K. E. J. Lehtinen, and M. R. Zachariah. Modeling and simulation of nanoparticle coagulation in high reynolds number incompressible flows. In *Proc. Joint US Sections Meeting of the Combustion Institute*, Oakland, CA, 2001.

S. L. Girshick and C. Chiu. Kinetic nucleation theory: A new expression for the rate of homogeneous nucleation from an ideal supersaturated vapor. *J. Chem. Phys.*, 93(2):1273–1277, 1990.

K. Kadoya, N. Matsunaga, and A. Nagashima. Viscosity and thermal conductivity of dry air in the gaseous phase. *J. Phys. Chem. Ref. Data*, 14(4):947–970, 1985.

M. Khakpour and S. C. Garrick. Effects of diffusion on nanoparticle coagulation and growth. In *Proc. Joint US Sections Meeting of the Combustion Institute*, Chicago, IL, 2003.

J. Ku, D. Griffin, P. Greenberg, and J. Roma. Buoyancy-induced differences in soot morphology. *Combust. Flame*, 102:216–218, 1995.

B. Liu, K. Rubow, P. McMurry, T. Kotz, and D. Russo. Airborne particulate matter and spacecraft internal environments. SAE Technical Paper 911476, 21st International Conference on Environmental Systems, San Francisco, CA, July 15-18 1991.

S. Modem and S. C. Garrick. Nanoparticle coagulation in a temporal mixing layer: mean and size-selected images. *J. Visualization*, 6(3):333–342, 2003.

G. Mulholland, M. Meyer, D. Urban, G. Ruff, Z. Yuan, V.Bryg, T. Cleary, and J. Yang. Pyrolysis smoke generated under low-gravity conditions. *Aerosol Sci. Technol.*, (49):310–321, 2015.

H. V. Nguyen, K. Okuyama, T. Mimura, Y. Kousaka, R. Flagan, and S. Seinfeld. Homogeneous and heterogeneous nucleation in a laminar flow aerosol generator. *J. Colloid Interf. Sci.*, 119:491–504, 1987.

E. Otto, H. Fissan, S. Park, and K. Lee. The Log-Normal Size Distribution Theory of Brownian Aerosol Coagulation for the Entire Particle Size Range: Part II – Analytical Solution Using Dahneke's Coagulation Kernel. *J. Aerosol Sci.*, 30(1):17–34, 1999.

T. Poinsot and S. Lele. Boundary conditions for direct simulations of compressible viscous flows. *J. Comp. Phys.*, 101:104–129, 1992.

A. Prakash, A. Bapat, and M. Zachariah. A simple numerical algorithm and software for solution of nucleation, surface growth, and coagulation problems. *Aerosol Sci. Technol.*, (37):892–898, 2003.

J. Pyykönen and J. Jokiniemi. Computational fluid dynamics based sectional aerosol modelling schemes. *J. Aerosol Sci.*, 31(5):531–550, 2000.

L. Talbot, R. Cheng, R. Schefer, and D. Willis. Thermophoresis of particles in a heated boundary layer. *J. Fluid. Mech.*, (101):737–758, 1980.

D. Urban, Z.-G. Yuan, P. Sunderland, G. Linteris, J. Voss, K.-C. Lin, Z. Dai, D. Sun, and G. Faeth. Structure and Soot Properties of Nonbuoyant Ethylene/Air Laminar Jet Diffusion Flames. *AIAA Journal*, 36 (8):1346–1360, 1998.

D. Urban, D. Griffin, G. Ruff, T. Cleary, J. Yang, G. Mulholland, and Z. Yuan. Detection of Smoke from Microgravity Fires. In *International Conference On Environmental Systems, 11 July*. SAE International, 2005.

N. Vargaftik, L. Filippov, A. Tarzimanov, and E. Totskii. *Handbook of Thermal Conductivity of Liquids and Gases*. CRC Press, 1993.

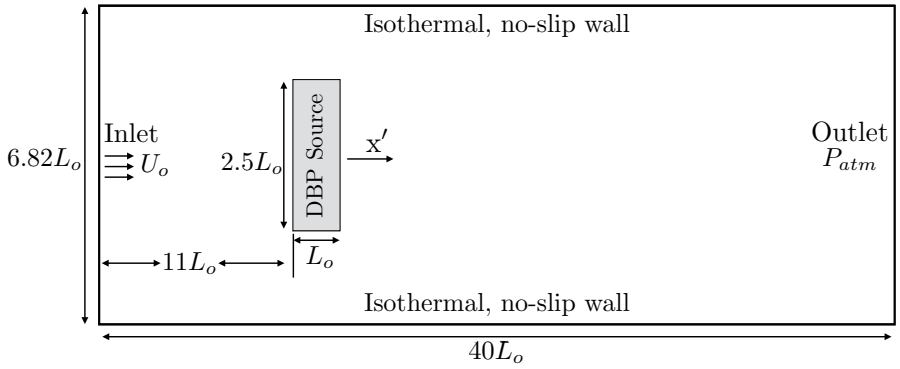


Figure 1: Flow geometry (not to scale).

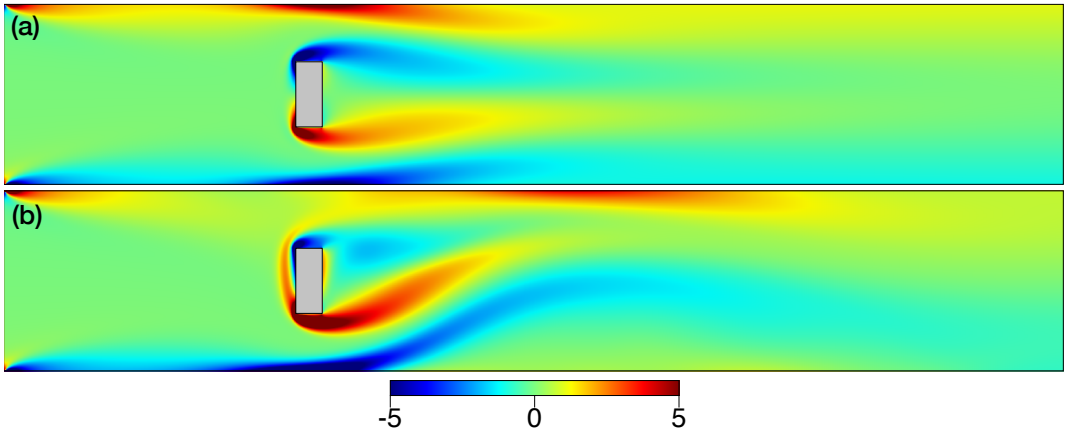


Figure 2: Dimensionless vorticity fields for (a) Case 1MG and (b) Case 1EG.

Table 1: Constant fluid properties.

Property	Air	Dibutyl phthalate
C_p	1007 J/Kg · k	2329.7 J/Kg · k
C_v	720 J/Kg · k	2299.83 J/Kg · k
MW	28.97 kg/kmol	278.3435 kg/kmol
R	287 J/kg · k	29.87 J/kg · k
γ	1.4	1.013
h^f	0 J/kg	-2465.7×10^3 J/kg

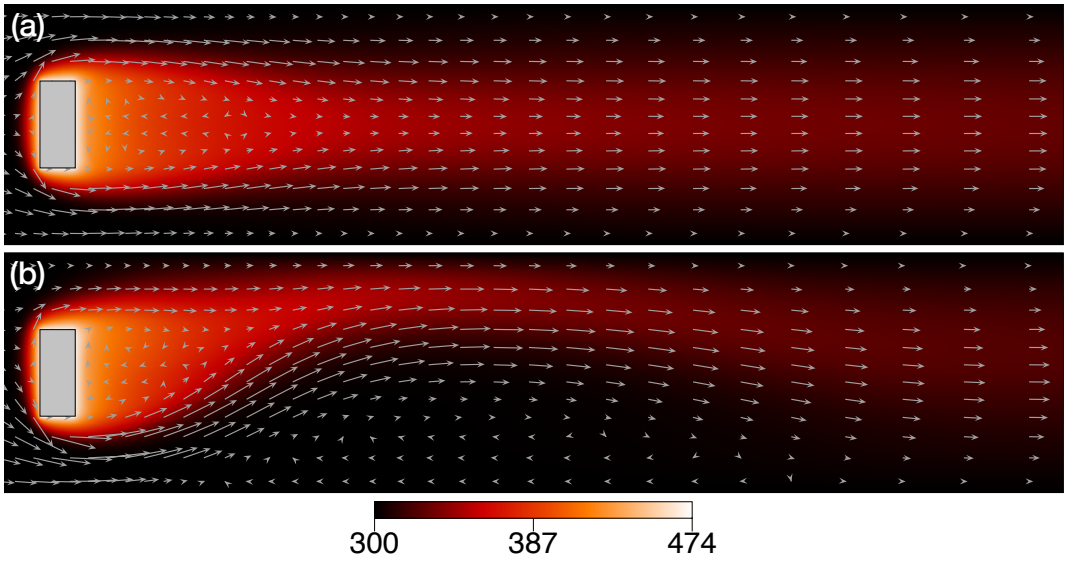


Figure 3: Temperature fields in Kelvin for (a) Case 1MG and (b) Case 1EG.

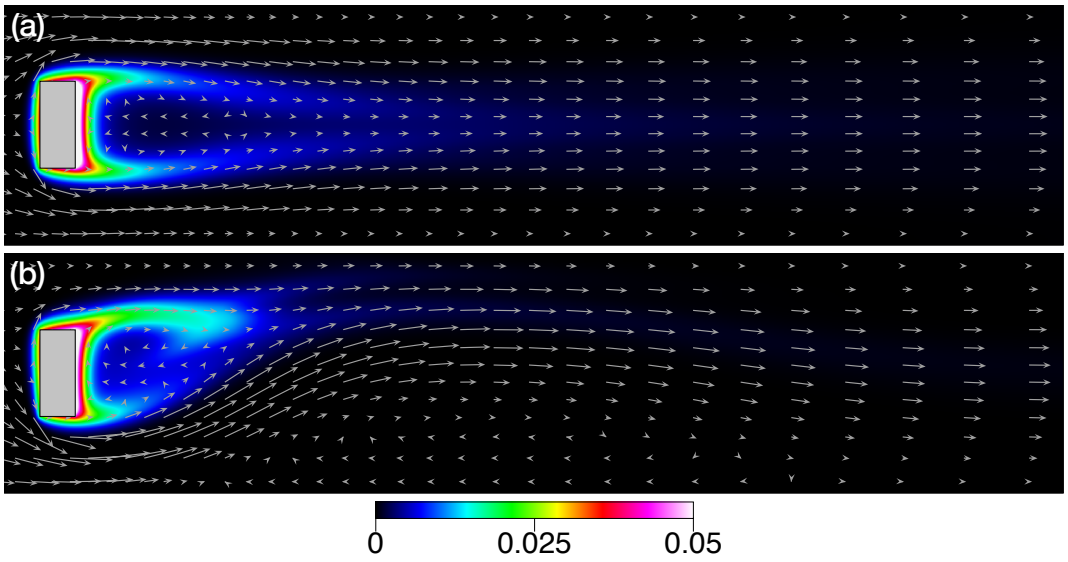


Figure 4: DBP vapor mass fraction fields for (a) Case 1MG and (b) Case 1EG.

Table 2: Variable DBP properties (Nguyen et al., 1987, Vargaftik et al., 1993).

Property	Functional form
Saturation pressure	$P_s = 133 \times \exp\left(16.27 - \frac{5099}{T-109.51}\right)\,[\mathrm{Pa}]$
Surface tension	$\sigma = 0.0353 - 0.0000863 \times (T - 273.16)\,\left[\mathrm{kg/s^2}\right]$
Density	$\rho_c = 1063 - 0.826 \times (T - 273.16)\,\left[\mathrm{kg/m^3}\right]$
Thermal conductivity	$k_p = 0.16738 - 0.00010357 \times T\,[\mathrm{W/m \cdot K}]$

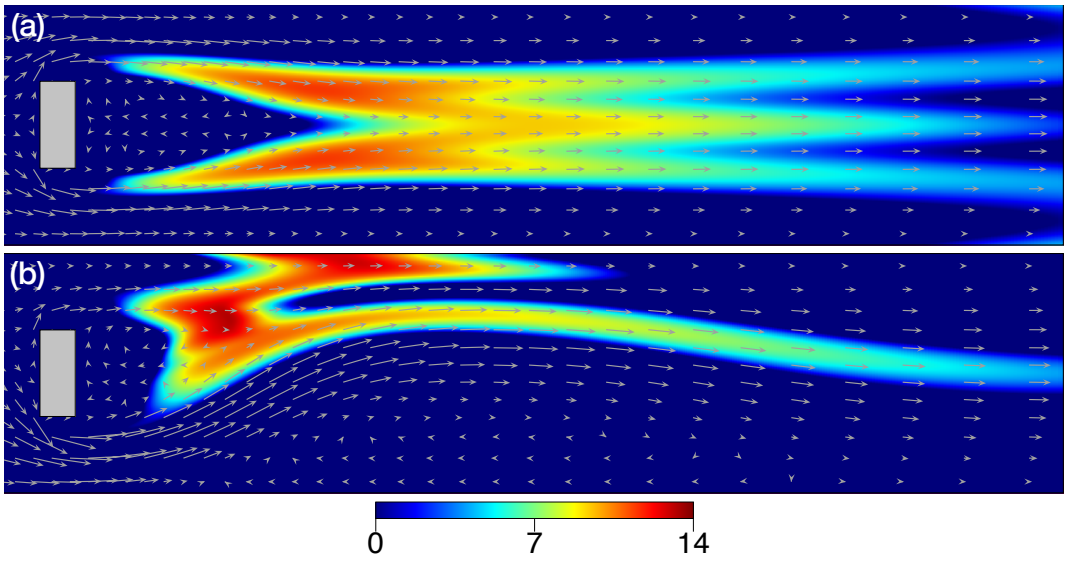


Figure 5: Log of particle nucleation rate $[m^{-3} \cdot s]$ for (a) Case 1MG and (b) Case 1EG.

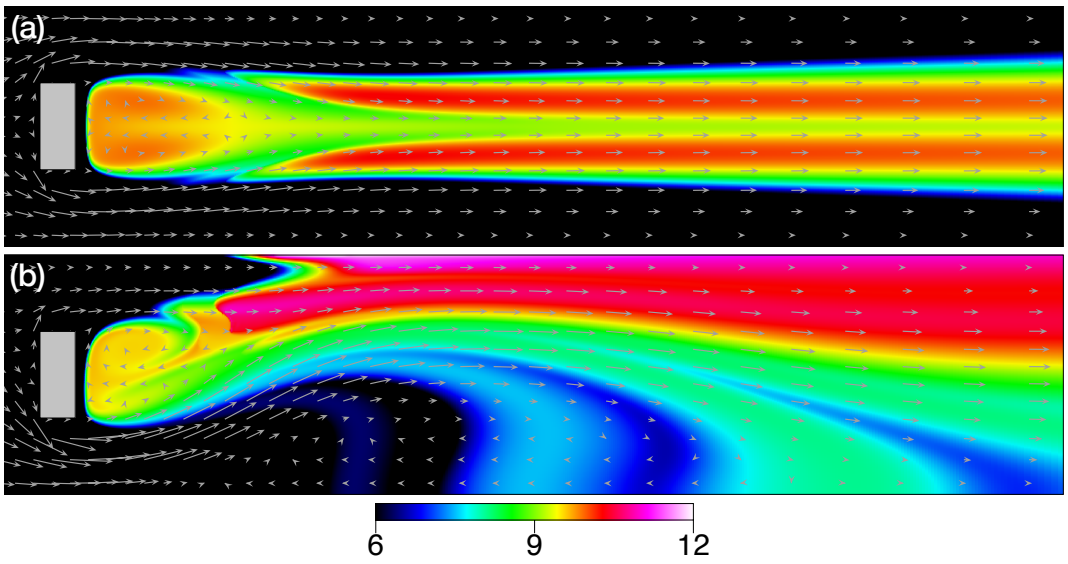


Figure 6: Log of particle number density $[m^{-3}]$ for (a) Case 1MG and (b) Case 1EG.

Table 3: Definitions of primary test cases.

Case	U_o[m/s]	g_o[m/s^2]	Re	Fr
1EG	0.08	9.81	22.3	0.385
1MG	0.08	0.0	22.3	∞
2EG	0.10	9.81	27.7	0.481
2MG	0.10	0.0	27.7	∞

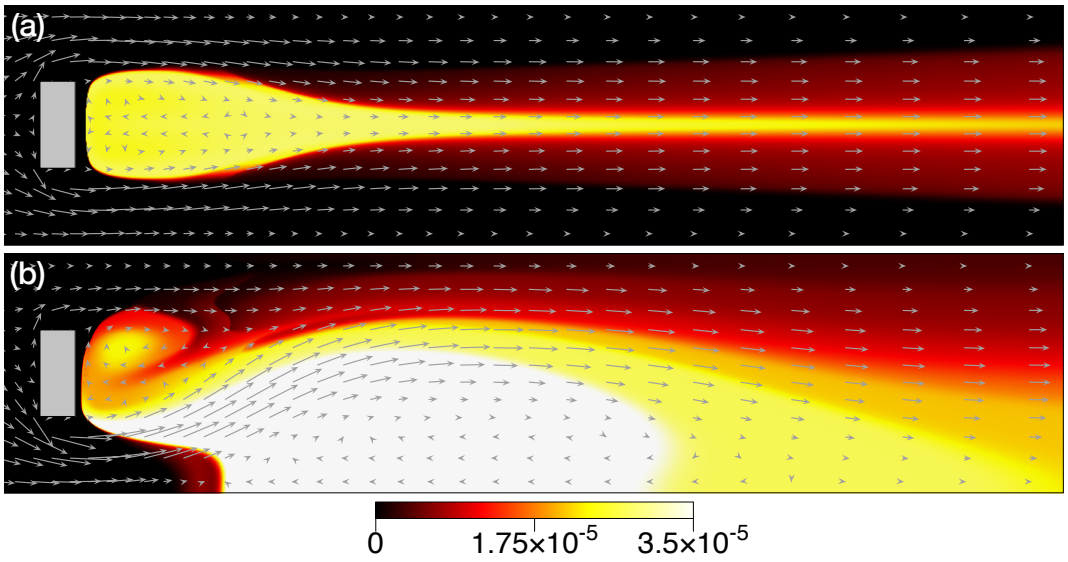


Figure 7: Diameter of average mass d_m $[m]$ for (a) Case 1MG and (b) Case 1EG.

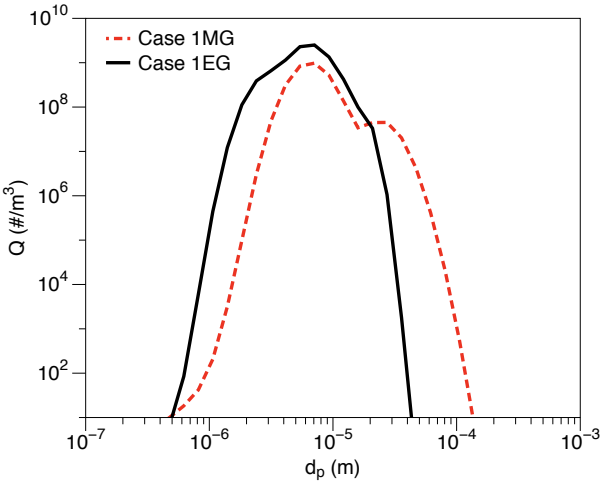


Figure 8: Particle size distributions for Case 1MG and Case 1EG computed from samples collected at $x' = 26$.

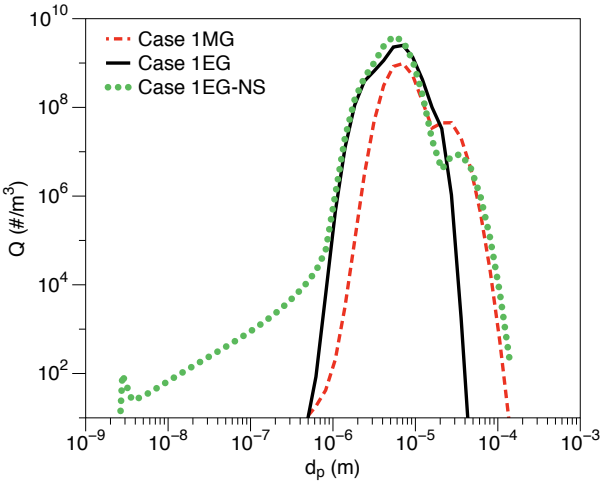


Figure 9: Particle size distributions for Case 1MG, Case 1EG, and Case 1EG–NS computed from samples collected at $x' = 26$

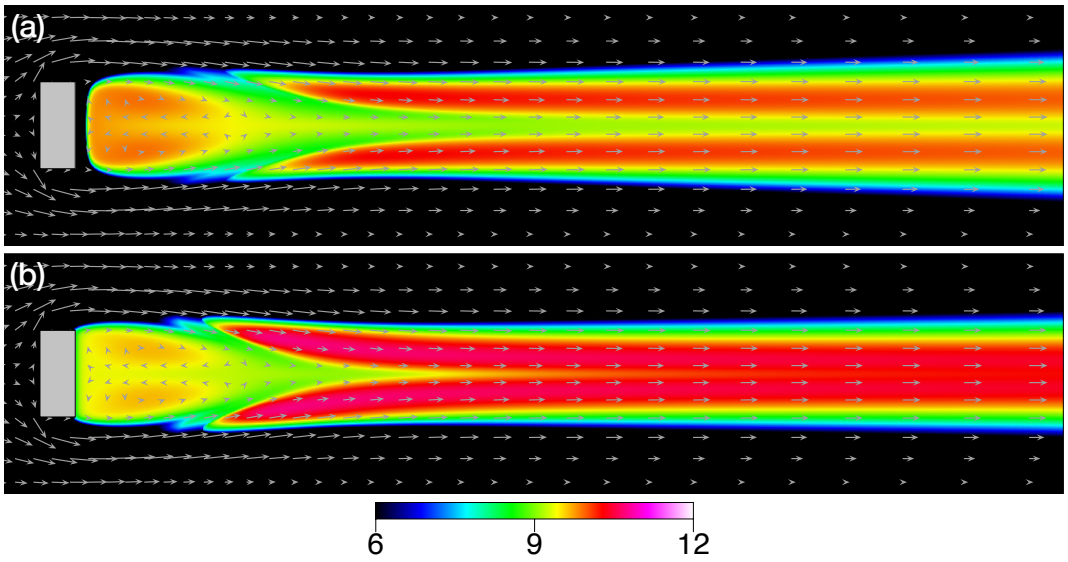


Figure 10: Log of particle number density $[m^{-3}]$ for (a) Case 1MG and (b) Case 1MG-NT.

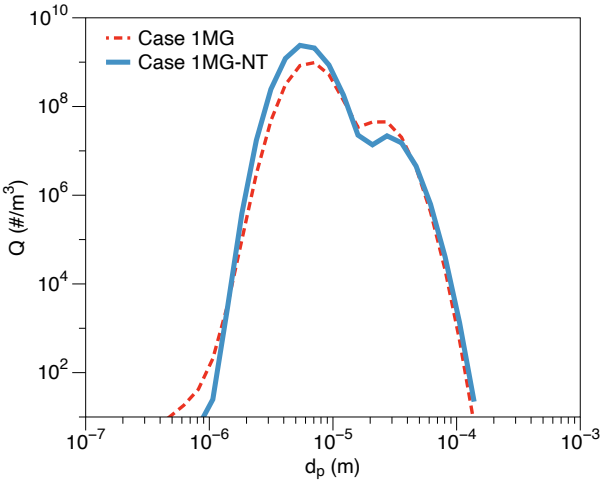


Figure 11: Particle size distributions for Case 1MG and Case 1MG-NT computed from samples collected at $x' = 26$.

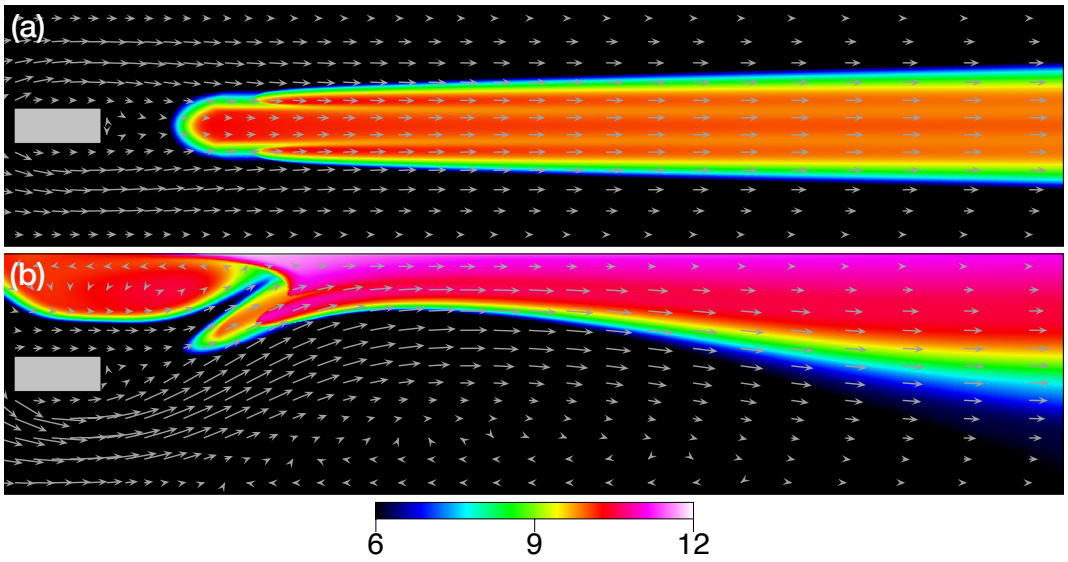


Figure 12: Log of particle number density $[m^{-3}]$ for (a) Case 1MG-R and (b) Case 1EG-R.

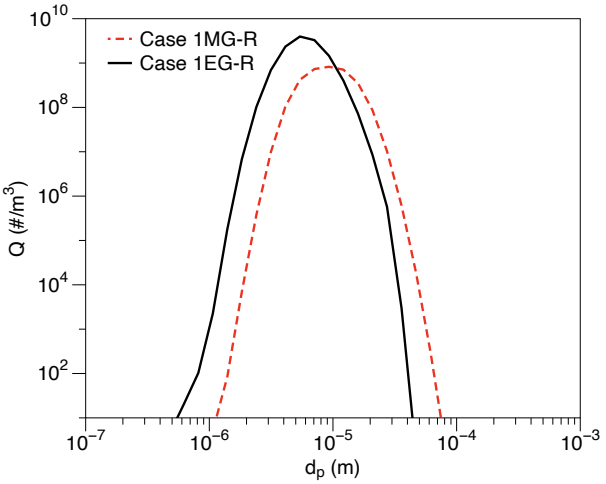


Figure 13: Particle size distributions for Case 1MG-R and Case 1EG-R computed from samples collected at $x' = 26$.

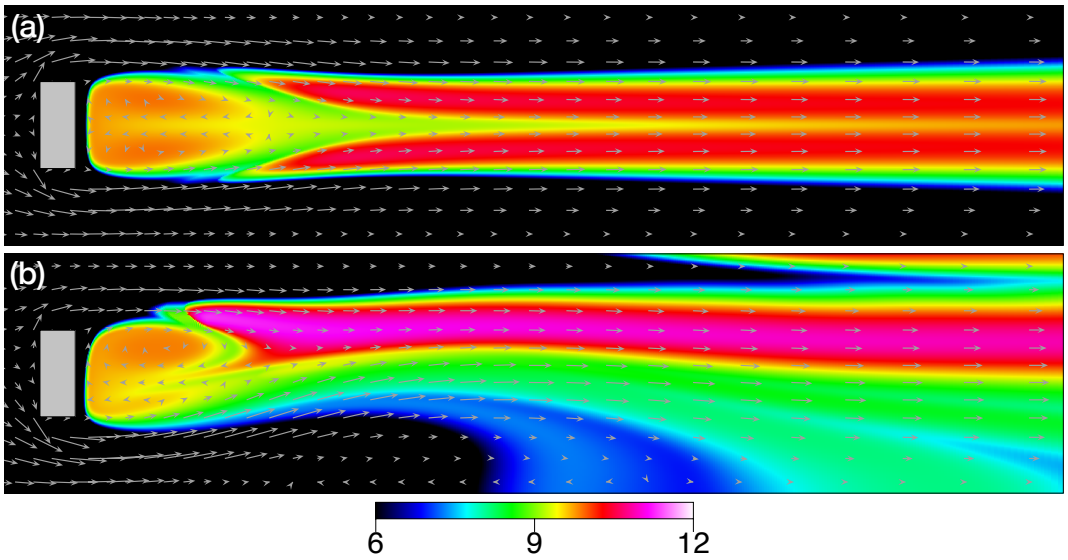


Figure 14: Log of particle number density $[m^{-3}]$ for (a) Case 2MG and (b) Case 2EG.

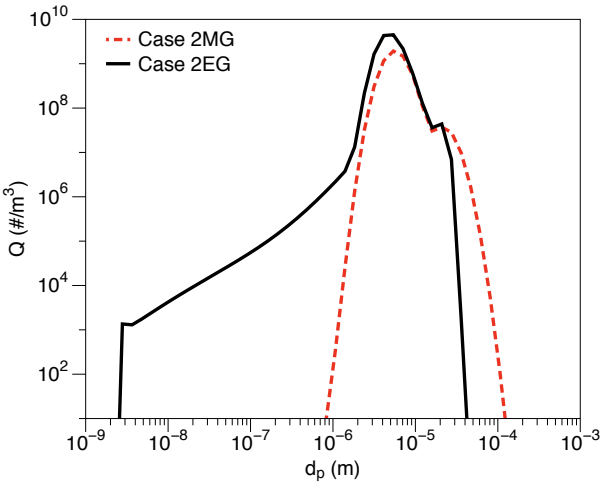


Figure 15: Particle size distributions for Case 2MG and Case 2EG computed from samples collected at $x' = 26$.

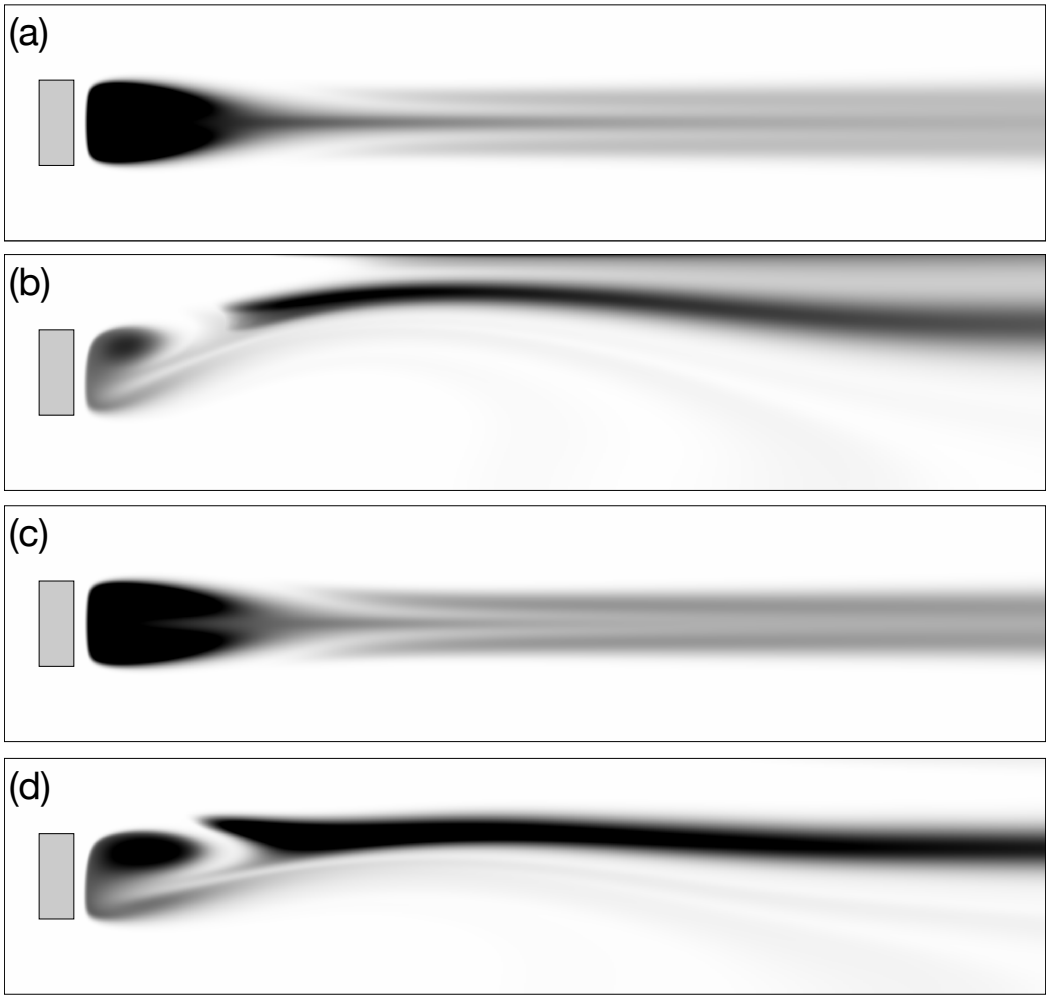


Figure 16: Visualization of simulated smoke fields for (a) Case 1MG, (b) Case 1EG, (c) Case 2MG, and (d) Case 2EG.

Table 4: Definitions of modified test cases.

Case	Modification type
1EG-NS	Case 1EG with particle settling deactivated
1MG-NT	Case 1MG with thermophoresis deactivated
1MG-R	Case 1MG with heated vapor source rotated 90°
1EG-R	Case 1EG with heated vapor source rotated 90°